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Original Article

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KEYWORDS
Deep learning;
Multitask detection;
Periapical radiograph;
Root canal filling

Abstract *Background/purpose:* Accurate evaluation of root canal filling (RCF) quality is paramount for successful endodontic outcomes. However, conventional radiographic interpretation is often subject to subjective variability. This study developed an automated, multitask artificial intelligence (AI) framework using YOLO11 and advanced image pre-processing to identify multiple dental structures and provide quantitative assessments of RCF quality on periapical radiographs.

Materials and methods: We proposed an image enhancement pipeline, which integrate intensity normalization, white top-hat transform, and linear contrast stretching to refine dental contours for training the YOLO11 multitask model. The framework simultaneously detected

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primary, permanent, dental crowns, and RCFs. Quantitative metrics, including root canal filling length (RCFL) and filling rate (RCFR), were calculated using pixel-accumulation and landmark localization algorithms.

Results: The proposed image enhancement significantly improved model performance, with the total mAP 50 increasing from 92.3% to 99.6%. The model achieved excellent diagnostic reliability, with intraclass correlation coefficients exceeding 0.96 across all tasks, notably reaching 0.997 for primary teeth. In clinical validation, the AI-assisted framework demonstrated high diagnostic agreement with the consensus of four dentists in classifying RCFL as adequate or underfilled. Furthermore, the AI-calculated RCFR for permanent (73.96%) and primary teeth (77.44%) showed high consistency with expert observations, while AI confidence levels consistently exceeded 95%.

Conclusion: The integrated YOLO11 framework provides a highly accurate, objective, and automated system for the quantitative evaluation of root canal obturation. By delivering reliable second opinions and precise apical limit localization, this system has strong potential to enhance diagnostic consistency and efficiency in clinical endodontic practice.

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Introduction

Endodontic treatment, commonly referred to as root canal treatment (RCT), is a fundamental dental procedure aimed at treating pathological conditions affecting the dental pulp and root canal system while preserving the natural tooth structure.[1] By removing infected or necrotic pulp tissue, disinfecting the canal system, and achieving adequate obturation, RCT effectively alleviates pain, eliminates infection, and maintains natural dentition. A systematic review has reported that more than 55% of the global adult population has at least one root-filled tooth (RFT), reflecting the widespread utilization of this procedure in contemporary dental practice.[2] Despite anatomical variations in the apical region, establishing the working length at approximately 1 mm short of the radiographic apex remains a clinically accepted standard when radiographs are used as a reference.[3] Inadequate obturation, particularly underfilling, may reflect incomplete canal debridement, residual necrotic tissue, and persistent bacterial infection. Meta-analyses have consistently demonstrated that the adequate extent of root canal filling (RCF) is a critical determinant of treatment outcomes.[3,4] When underfilling is identified, close follow-up or retreatment should be considered prior to definitive prosthetic rehabilitation.

While sharing the same fundamental goal of infection control, the clinical considerations of RCT differ between permanent and primary teeth. In deciduous dentition, treatment strategies emphasize minimizing damage to the developing permanent successors; therefore, instrumentation is generally recommended to terminate 1—2 mm short of the radiographic apex. Given these differences, accurate assessment of root canal filling quality is essential in both dentitions.[5,6] Periapical radiographs (PAs) are widely used in clinical practice for evaluating root canal fillings and identifying dental structures, including permanent teeth and prosthetic restorations such as dental crowns. However, interpretation of radiographic findings largely depends on clinicians' subjective judgment and experience, which may result in inter-observer variability and inconsistent diagnostic outcomes.[7—10]

With the rapid advancement of artificial intelligence (AI), deep learning (DL) techniques have demonstrated strong potential in dental radiographic analysis.[11,12] Convolutional neural networks (CNNs) and related DL models have been successfully applied to tasks such as tooth detection, structural identification, and disease classification, improving diagnostic accuracy and clinical efficiency.[13—15] However, relatively limited research has focused on the automated and quantitative evaluation of root canal filling quality, particularly in conjunction with permanent tooth identification and detection of prosthetic restorations within a unified analytical framework on periapical radiographs.[16—18] Therefore, the aim of this study was to develop an AI-assisted system for automated analysis of PA radiographs, enabling detection of root canal filling materials in both permanent and primary teeth, classification of filling extent, identification of permanent teeth, and detection of dental crowns, with the goal of improving diagnostic consistency and supporting clinical decisions-making.

Materials and methods

PA dataset collection and annotation

This study used a PA dataset collected from 1060 patients, comprising 492 adults and 568 children. The research protocol received formal approval from the Institutional Review Board (IRB) of the chang gung medical foundation (IRB number: 202400084B0), chang gung memorial hospital, Taoyuan, Taiwan. All PAs were captured using a high-frequency X-ray generator (HELIODENT PLUS, Sirona Dental Systems GmbH, Bensheim, Germany) configured to deliver

a constant high voltage. The exposure time was fine-tuned within a range of 0.03—3.2 s, tailored to specific clinical requirements. Digital sensors with physical dimensions of 31.3 × 44.5 mm were utilized, generating images with resolutions of either 825 × 1200 or 820 × 562 pixels. All acquired images were systematically preserved in DCI format for subsequent analysis. Regarding the dataset partition, 160 PAs were reserved as an independent test set for final model evaluation. The remaining images were divided into a training set (80%) and a validation set (20%). To ensure high-quality ground truth, all images were annotated by four oral medicine specialists, each with at least 3 years of clinical experience. Furthermore, all research procedures were conducted under the rigorous supervision of senior AI experts and pediatric dentists to mitigate operator-related bias and ensure clinical relevance.

PA pre-processing

This study proposes an advanced pre-processing framework for PAs to enhance the performance of deep learning models in multitask diagnostic challenges. First, all raw PAs (Fig. 1A) were size-normalized to a standardized resolution of 640 × 640 pixels (Fig. 1B) to optimize the computational efficiency of AI-assisted detection. Subsequently, intensity normalization was applied to the training dataset to ensure uniform brightness and contrast (Fig. 1C). To highlight critical anatomical features, a white top-hat transform was used to extract dental contours (Fig. 1D). This technique, in combination with intensity normalization, effectively suppressed large-scale grayscale fluctuations. The extracted contours were then refined using linear contrast stretching to enhance edge prominence (Fig. 1E). Finally, these enhanced features were overlaid onto the normalized images (Fig. 1B) to produce the final training input (Fig. 1F). This integrated pipeline strengthens the representation of key dental structures, providing more robust data for model training.

YOLO detection

This study presents an intelligent dental image analysis framework that integrates you only look once (YOLO) with advanced image pre-processing techniques. The framework enables the automated detection and classification of dental structures on PAs, including primary teeth, permanent teeth, dental crowns, and root canal fillings, while providing quantitative support for root canal treatment quality assessment. The core of this system, YOLO11 (Ultralytics, Washington, D.C., USA, 2024) is a state-of-the-art multitask deep learning architecture. It features an enhanced C3k2 backbone for the robust extraction of high-density anatomical features and a PAN-FPN neck for effective multi-scale feature integration. By incorporating a decoupled detection head, the model enables the simultaneous execution of multiple object detection tasks, thereby enhancing both efficiency and objectivity in clinical dental diagnostics.

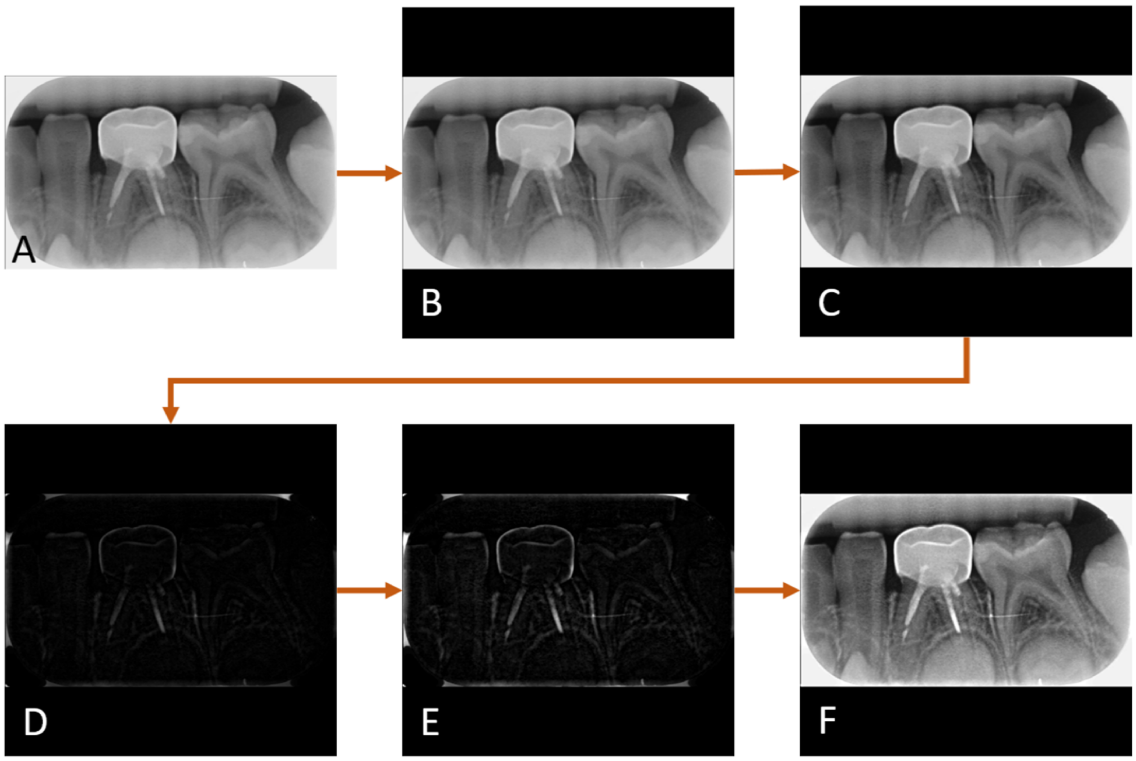


Figure 1 Periapical radiograph (PA) pre-processing with (A) original PA (B) size normalization (C) intensity normalization (D) white top-hat transform (E) linear contrast stretching, and (F) overlay PA.

Root canal filling length evaluation method

To evaluate root canal filling length (RCFL) and root canal filling rate (RCFR), the YOLO11-seg model was employed to perform precise segmentation of the RCF (Fig. 2B1) and the tooth structure (Fig. 2B2). Subsequently, a pixel-accumulation algorithm was used to identify the lowest coordinates of both segmented regions, which were designated as the root canal filling bottom point (Fig. 2C1) and the root apex point (Fig. 2C2), respectively. Finally, these identified keypoints were overlaid to generate composite visualization (Fig. 2D) for clinical assessment. In this study, RCFL was categorized into underfilled and adequate; cases of overfilling were excluded from the current scope of discussion. For RCFL measurement, a two-point distance evaluation was performed using (1), where (x_f, y_f) represents the pixel coordinates of the root filling endpoint, and (x_f, y_f) denotes the pixel coordinates of the root apex. Regarding the evaluation of RCFR, the adequacy of the filling was assessed via (2), where M_{final} represents the number of pixels in the final segmented filling region. To establish an objective ground truth independent of the AI predictions, the ideal root canal mask (M_{ideal}) was manually delineated by senior dentists.

$$RFL = \sqrt{(x_f - x_a)^2 + (y_f - y_a)^2} \quad (1)$$

$$RCFR(\%) = \frac{|M_{final}|}{|M_{ideal}|} \times 100\% \quad (2)$$

Framework evaluates metrics

To rigorously quantify the diagnostic precision and reliability of the proposed multitask model, a suite of standardized evaluation metrics was employed. As detailed in (3)–(6), accuracy, precision, sensitivity (recall), F1-score, and mAP 50 were derived from the confusion matrix. Furthermore, the model's diagnostic performance and tooth detection consistency were validated through the receiver operating characteristic (ROC) curve, area under the curve (AUC), and the intraclass correlation coefficient (ICC).

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (4)$$

$$F1 - score = \frac{2 \times Precision \times Sensitivity}{Precision + Sensitivity} \quad (5)$$

$$mAP50 = \frac{1}{n}\sum_{i=1}^{n} AP_{i,iou=0.50} \quad (6)$$

Results

YOLO11 multitask segmentation evaluation

This study first evaluates the efficacy of the proposed image enhancement (PP) method and performs comparative analysis against the original (ORI) dataset. As shown in

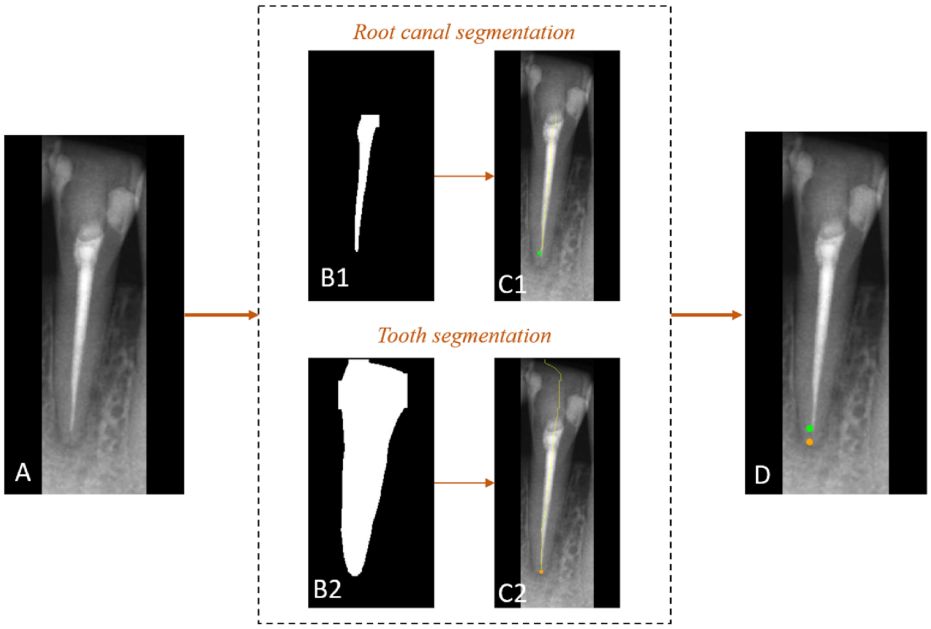


Figure 2 Root canal treatment evaluation with (A) root canal tooth segment result from YOLO11 (B1) root canal filling segmentation from root canal tooth (B2) tooth segmentation from root canal tooth (C1) filling bottom localization (green point) (C2) root apex localization (orange localization) and (D) overlay green point and orange point to measure distance and filling rate.

Table 1, implementing PP significantly improved the model's performance across all dental classifications. Precision and mAP 50 increased from 93.9% to 92.3%—97.9% and 99.6%. The detection of dental crowns showed a substantial improvement in mAP50, rising from 92.2% to 97.2%, demonstrating that the PP method effectively enhances the model's ability to distinguish complex dental structures and restorations. The diagnostic efficacy and reliability of the proposed YOLO11 multitask model are shown in Fig. 3. The ROC analysis (Fig. 3A) demonstrated superior discriminative power, with AUC values reaching 0.998 for crowns and exceeding 0.89 for primary teeth, permanent teeth, and RCFs. This high diagnostic accuracy was further confirmed by the confusion matrix (Fig. 3B), which showed minimal misclassification across all dental categories. Furthermore, the ICC analysis (Fig. 4) revealed an excellent consistency (ICC $>$0.96) between YOLO11 predictions and ground truth across all tasks. Notably, the ICC for primary teeth reached 0.997, indicating that the model provides highly objective and stable measurements that are clinically comparable to dentist's annotations.

Comparison with dentist's ground truth and proposed AI-assisted framework

In this study, YOLO11-based detection results served as the primary basis for the subsequent quantitative evaluation of RCFL and RCFR. Table 2 presents a comparative analysis of the AI-assisted framework and four dentists, comprising two senior dentists (attending physicians with over seven years of clinical experience) and two junior dentists (resident physicians with at least three years of clinical experience). To ensure diagnostic objectivity, all dentists performed their evaluations independently without interference, and their findings were averaged to establish the "Dentist's evaluation and confidence" baseline.

In the RCFL assessment, the AI0-assisted framework achieved high diagnostic agreement with the clinical team regarding both underfilled and adequate categories. Notably, following the stratified analysis, the AI's confidence levels (ranging from 95.89% to 97.23%) were demonstrated to be highly comparable to those of the senior dentists (96.42%—97.63%) and consistently higher than those of the junior dentists (91.80%—93.25%). Furthermore, in the quantitative analysis of RCFR. Particularly for primary teeth, the AI framework-calculated value (77.44%) closely approximated the senior group's evaluation (79.95%), significantly outperforming the junior group (65.44%). For permanent teeth, the AI's calculation (73.96%) also aligned well with clinical observations, exceeding the junior group's confidence (72.91%). These findings strongly validate the framework's utility as a reliable professional second opinion.

Table 1 Evaluation metrics for disease detection using YOLO11.

Metrics	Precision		Recall		F1-score		mAP 50	
Classification	ORI	PP	ORI	PP	ORI	PP	ORI	PP
Primary teeth	92.9%	97.03%	92.6%	95.6%	91.95%	96.71%	95.8%	97.4%
Permanent teeth	93.9%	97.93%	92.6%	95.6%	92.72%	96.88%	92.8%	97.4%
Root canal filling	93.4%	97.04%	91.9%	94.9%	91.36%	96.19%	91.6%	96.2%
Dental crown	89.2%	91.41%	91.3%	96.7%	92.42%	94.21%	92.2%	97.2%
Average	93.9%	97.9%	91.9%	99.9%	92.77%	98.23%	92.3%	99.6%

ORI, original PA; PP, pre-processed PA; mAP 50, Mean average precision at IoU threshold of 0.5.

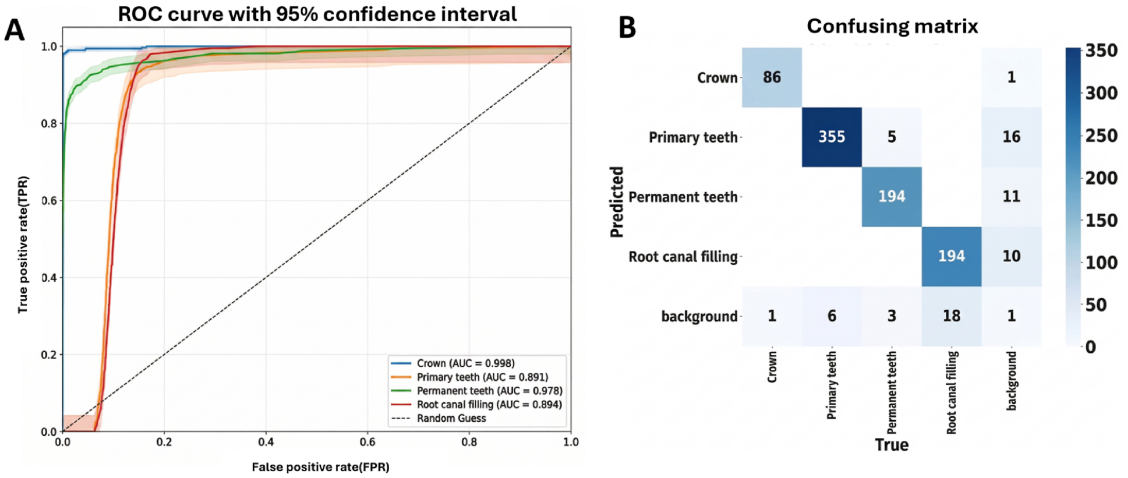


Figure 3 Comprehensive performance evaluation of the YOLO11 multitasks (A) ROC curves and AUC values for different dental classifications (B) Confusion matrix displaying the predicted vs. ground truth categories.

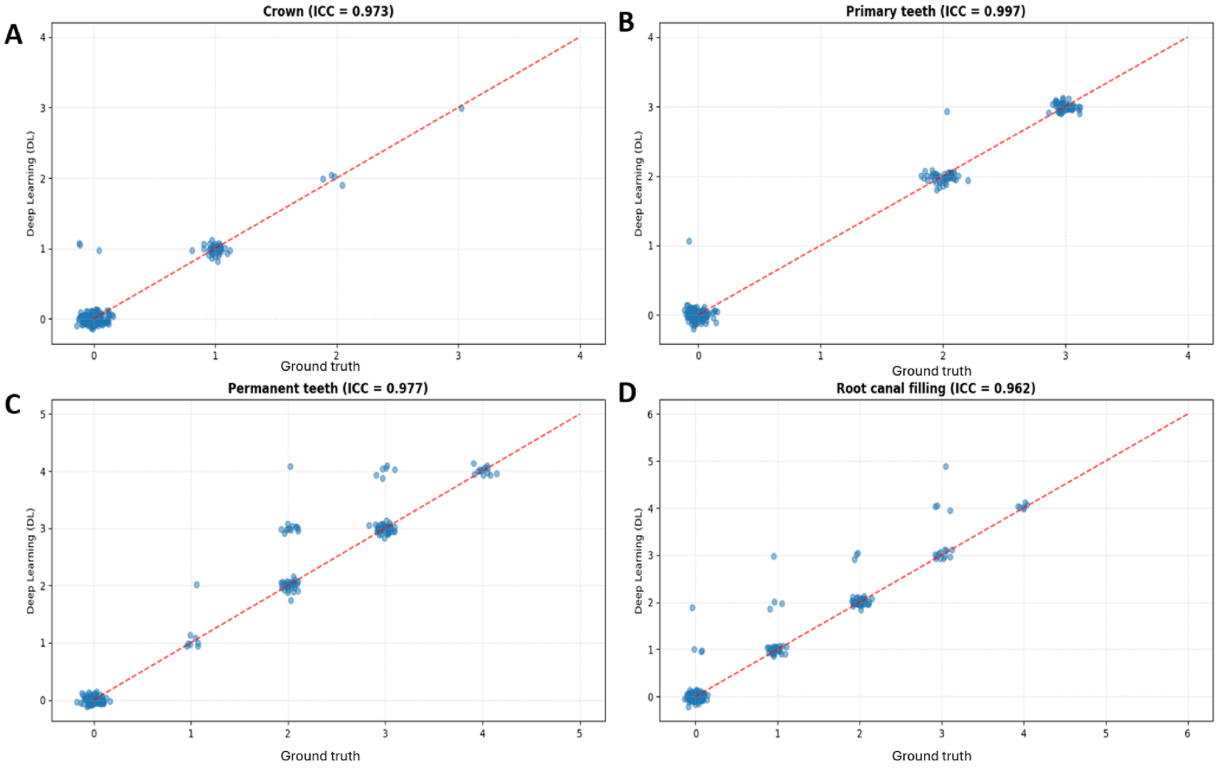


Figure 4 Correlation plots and ICC between YOLO11 predictions and ground truth for (A) crowns (B) primary teeth (C) permanent teeth and (D) root canal fillings.

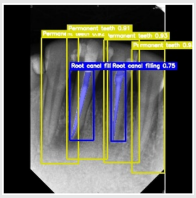
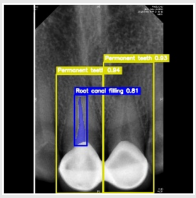
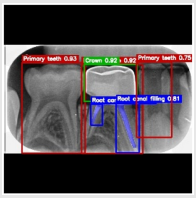
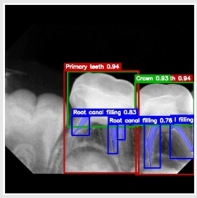
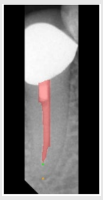
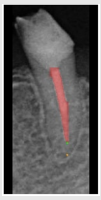
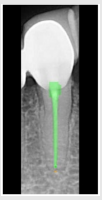
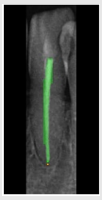
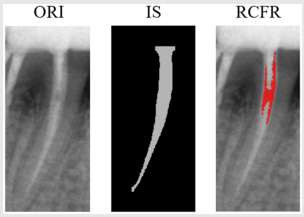
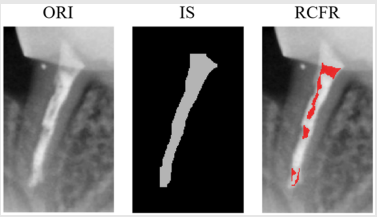
Discussion

Apical periodontitis is a complex and severe biological complication that requires accurate endodontic diagnosis and treatment to prevent tooth loss and ensure timely, evidence-based healing.[19] Tailored treatment plans should incorporate precise mechanical instrumentation and obturation, considering the root canal morphology and the apical limit of the filling to optimize clinical outcomes and prolong tooth longevity.[20] Currently, PAs are commonly used in clinical practice to assess the quality of RCF.[21] The adequacy of the obturation is typically estimated by visually evaluating the relative distance between the apical point of the RCF and the radiographic apex.[22] However, variability in image quality, superimposition of anatomical structures, and unclear apical landmarks often contribute to difficulties in accurate measurement and to inconsistencies arising from subjective visual interpretation. In clinical settings involving a high volume of endodontic cases or limited access to specialized expertise, particularly in primary care clinics, there is an urgent need to establish a standardized, objective, and automated diagnostic support system to improve diagnostic efficiency and consistency.

Recent DL models have demonstrated high reliability in accurately analyzing various dental anatomical features and restorations on PAs, achieving over 95% classification accuracy in tooth and RCF identification tasks.[23] However, to our knowledge, limited comprehensive studies have addressed the fully automated, quantitative evaluation of root canal obturation quality, specifically the precise location of the apical limit. For example, Szabó et al. proposed an initial detection approach using a CNN-based model to classify the presence of RCF,[24] reporting an accuracy of 89%, though without assessing the clinical adequacy of the obturation. Subsequently, Issa et al. built upon this foundation by integrating advanced object detection networks to improve the screening efficiency of endodontically treated teeth and achieve 92.30% sensitivity to detect RCF.[25] These methods have achieved high classification accuracy, representing a significant improvement over traditional manual evaluation. Nevertheless, most of these studies primarily focused on identifying the mere existence of RCF on a macroscopic level rather than quantifying the exact distance to the radiographic apex or assigning clinical quality levels, such as underfilling. To address this, contemporary models have attempted to evaluate RCF quality using bounding box predictions to estimate the spatial relationship between the filling and the root apex.[26] However, this method presents two key limitations: first, the bounding box coordinates defined by general object detection models lack the pinpoint precision required for microscopic endodontic measurements, resulting in variation across repeated inferences; second, the precise localization of the filling's apical point compared to the true radiographic apex remains a partially manual task, failing to achieve full automation and still imposing a diagnostic burden on clinical practice.

In contrast, our study aimed to address the limitations encountered by previous state-of-the-art methods in endodontic image analysis. By integrating DL-based localization with advanced image processing techniques for precise

Table 2 Representative segmentation results and objective quantification of root canal filling length (RCFL) and root canal filling rate (RCFR) using the proposed AI-assisted framework.

YOLO11 segmentation result				
Root canal filling length (RCFL) evaluation				
RCFL visualization				
Average dentist's confidence	Underfilled /94.25%	Underfilled /93.75%	Adequate /93.50%	Adequate /93.75%
Senior dentist's group confidence	Underfilled /97.11%	Underfilled /96.42%	Adequate /97.22%	Adequate /97.63%
Junior dentist's group confidence	Underfilled /92.72%	Underfilled /92.99%	Adequate /91.80%	Adequate /93.25%
AI-assisted framework	Underfilled/97.23%	Underfilled/95.89%	Adequate /96.61%	Adequate /97.23%
Root canal filling rate (RCFR) evaluation				
Category	Permanent teeth		Primary teeth	
RCFR visualization				
Dentist's evaluation	Average confidence	75.42%	Average confidence	70.13%
	Senior confidence	81.66%	Senior confidence	79.95%
	Junior confidence	72.91%	Junior confidence	65.44%
AI-assisted framework	73.96%		77.44%	

macroscopic detection, our system achieved an overall mAP 50 of 99.6% and a sensitivity of 99.9%, representing a 7.3% and 8% improvement, over the baseline models using original images without preprocessing. Furthermore, instead of relying on imprecise bounding boxes, we established the exact pinpoint coordinates of the radiographic apex and the apical limit of the obturation as the definitive baseline for evaluating treatment quality. We developed a tailored landmark detection algorithm based on absolute distance calculations to accurately classify the clinical status of RCF and efficiently identify underfilling conditions. The key contributions and innovations of this study are summarized as follows.

1. The proposed image enhancement pipeline effectively suppressed grayscale fluctuations, resulting in substantial performance gains, with mAP 50 increasing from 92.3% to 99.6% compared to the baseline model.
2. The AI-assisted framework shown exceptional diagnostic results with an ICC exceeding 0.96 across all tasks, with primary teeth achieving an ICC of 0.997. Furthermore, the result attained high diagnostic agreement with the consensus of four dentists in classifying RCFL.
3. The AI-assisted framework established precise coordinates for the root apex and the filling endpoint, moving beyond imprecise bounding boxes. This objective approach provided quantitative RCFL and RCFR

data with confidence levels consistently exceeding 95%.

However, this study has several limitations. The AI-assisted framework proposed by this study was heavily dependent on the absolute distance calculated between the filling material and the radiographic apex. Based on our experimental findings, borderline cases, such as those where the apical point of the filling ends exactly near the critical clinical thresholds, presented significant measurement challenges. Our landmark detection model struggled to consistently and accurately classify cases in these extremely narrow borderline ranges, particularly in teeth with severe root curvature or complex apical morphologies, leading to decreased classification performance, with a sensitivity of only approximately 82.33%. In addition, we tested the model under suboptimal periapical image quality conditions, such as severe superimposition of anatomical structures or images with extreme brightness and noise. Under these circumstances, the precise localization and classification accuracy declined to approximately 69.14%—81.12%, accompanied by a noticeable reduction in the model's confidence scores for predicting the exact coordinates of the apical landmarks. Consequently, while the AI framework demonstrates a diagnostic consistency highly comparable to that of senior specialists in general scenarios, a performance gap still exists. The AI cannot yet fully replicate the nuanced, experience-based clinical judgment of senior dentists, particularly when interpreting these ambiguous or suboptimal radiographs. In future work, we aim to incorporate more diverse datasets with complex anatomical variations and explore the integration of advanced artifact-reduction algorithms to mitigate the impact of two-dimensional superimpositions and suboptimal imaging conditions.

In conclusion, this study has validated the integration of AI and advanced image processing techniques as a highly practical approach for accurately assessing RCF quality and precisely locating the apical limit. The proposed two-stage automated system demonstrated promising clinical outcomes, providing reliable, objective support for experienced dental professionals during routine endodontic evaluations and serving as a valuable educational tool for novice dentists. This research significantly advances automated, quantitative diagnostic technologies in clinical dentistry and underscores the practical value of AI-assisted tools in enhancing diagnostic accuracy, consistency, and overall treatment efficiency.

Declaration of competing interest

The authors have no conflicts of interest relevant to this article.

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